INCH-POUND
MIL-PRF-3098/85D
7 August 2009
SUPERSEDING
MIL-PRF-3098/85C
17 May 2004

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR109/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 10 MHz to 11.5 MHz; fundamental; controlled; antiresonance.

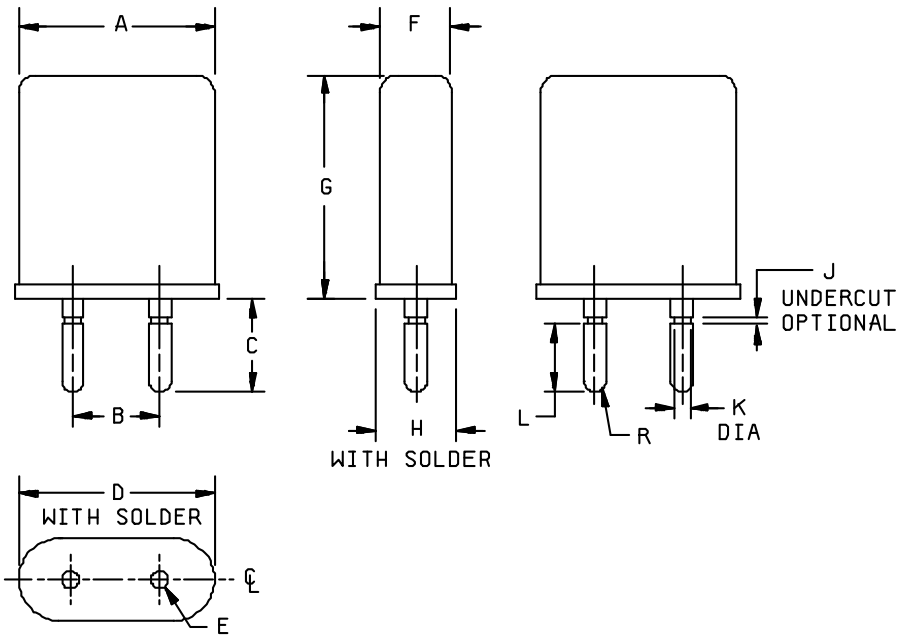

FIGURE 1. Crystal unit - CR109/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	.395	.405	10.03	10.29
B	.182	.202	4.62	5.13
C	.220	.280	5.59	7.11
D	.410	.430	10.41	10.92
E	.038	.042	0.96	1.67
F	.145	.155	3.68	3.94
G	.432	.470	10.97	11.94
H	.160	.180	4.06	4.57
J	.015	.025	0.38	0.64
K	.027	.032	0.68	0.81
L	.165	.175	4.19	4.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR109/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 10 MHz to 11.5 MHz, inclusive.

Frequency tolerances:

Operating temperature range: ±0.002 percent.

Reference temperature: ±0.002 percent.

Room temperature: ±0.008 percent.

Frequency stability: ±0.0005 percent.

Mode of oscillation: Fundamental.

Reference temperature: 85°C ±2°C.

Temperature ranges:

Operating (controlled): +80°C to +90°C, inclusive.

Operable: -55°C to +80°C, inclusive.

Capacitance, load: 32.0 pF ±0.5 pF.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Resistance: 11 ohms.

Crystal current: 15 mA.

Capacitance, shunt: 7 pF, maximum.

Equivalent resistance: 22 ohms, maximum.

Frequency change permitted: ±0.0004 percent.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201, MIL-STD-202; double amplitude .036 inches (0.91 mm) or 10 gravity units, whichever is the limiting value. Frequency shall be cycled between 10 Hz and 500 Hz in 15-minute cycles.

Frequency change permitted: ±0.0004 percent.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±0.0005 percent.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Air Force - 19, 84

Preparing activity:
DLA - CC

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